

Experimental and FEM Verification of Cogging Forces of a Permanent Magnet Linear Motor with a Discontinuous Magnet Track

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Abstract—Permanent magnet track linear motor drives offer a cost-effective motion control solution in numerous industrial manufacturing processes, covering a wide range of material handling and conveyor applications. To further reduce the initial costs of a permanent magnet track linear motor drive system, discontinuous magnet tracks can be considered. However, in a discontinuous magnet track, the armature has to face regions with and without permanent magnets, thereby resulting in edge crossings, where the armature is only partly overlapping with the permanent magnets. As a result, both the cogging force and electromechanical forces cause high disturbances at the edge crossings, having a major impact on the performance of the motion controllers. In this paper, disturbances of a nonsalient permanent magnet linear motor with a discontinuous magnet track are studied. An experimental test setup for the estimation of disturbance forces at the edge is presented. The behavior of the estimated disturbance forces at the edge crossings is also evaluated by a FEM model and compared with the experimental results.

Index Terms—Discontinuous magnet track, Cogging forces, Permanent magnet linear motor (PMLM).

I. INTRODUCTION

In traditional permanent magnet linear motors (PMLM), the moving part is the armature and the stationary part consists of permanent magnets. The excitation of the motor is achieved by approximately constant fluxes from permanent magnets, and if the armature contains ferromagnetic materials, such as iron, they will have a direct interaction with the permanent magnets. Because of slot openings, the permeance of the armature varies causing cogging forces with the excitation flux produced by the permanent magnets. As a result, the cogging forces will produce disturbances to the total force in the direction of motion. One of the most important design aspects of linear motors, from the viewpoint of precision machinery, is the compensation or minimization of the influence of cogging forces.

Methods to reduce cogging forces of rotor surface magnet permanent magnet machines, from the motor design perspective, have been presented in [1]. Although the analysis has been made for rotating motors, the same principles mainly apply also for linear surface magnet motors. In particular, modifications in the mechanical design of the linear motor,

including winding arrangements, significantly reduce the cogging forces as was shown in [2]–[4]. Correspondingly, the cogging force effects in linear motor drive can be reduced by using advanced control methods as shown in [5], [6].

The terms 'segmented' or 'discontinuous' are commonly used for describing a linear motor, which has a discontinuous magnet track. These tracks can be divided into tracks with short gaps and tracks with long gaps. In the case of a track with short gaps, the gaps are shorter than the length of the mover. This means that the mover overlaps at least two track elements all the time, like in [7]. Then again, when a track with long gaps is considered, the gaps are longer than the length of the mover. In other words, the track contains areas where the mover is not above the track element and is freewheeling. The principle of application with long gaps can be found for instance in [8]. In the case of discontinuous tracks, the influence of cogging forces is stronger at the edges of the armature segment than in a situation where the magnet plate is totally covered by the armature segment. The reason for this is an unbalance of the cogging forces during entry or ejecting intervals. An analysis of the cogging forces of the outlet edge of the armature segment is provided in [9], [10].

Even though the term 'discontinuous permanent magnet track' is applied in this paper and the experimental test setup under study has a moving armature, a common practical solution is a short mover with permanent magnets. In that case, the track elements consist of a segmented, discontinuous armature. The overlapping phenomena are naturally similar in both assemblies. This paper studies the disturbance forces of a PMLM at the magnet track edges of a discontinuous magnet track. It is shown by experimental measurements that the cogging forces produce a major part of the disturbance forces, whereas electromechanical forces caused by the edge have only a minor effect.

The contents of this paper are organized as follows. First, the problem statement is addressed in Section II by discussing the cogging forces of a PMLM in the case of a discontinuous magnet track. After that, a 2D-FEM-modeling of a nonsalient permanent magnet linear motor under study is introduced. Section III introduces an estimation routine of the disturbance

forces at the edge, and Section IV presents the test setup that is used to validate the behavior of the cogging forces during the edge crossings. The experimental results are analyzed and compared with the FEM calculation result. Section V concludes the paper.

II. PROBLEM STATEMENT

Permanent magnet track linear motor drives are used in numerous industrial manufacturing processes, covering a wide range of material handling and conveyor applications. To further reduce the initial costs of a permanent magnet track linear motor drive system, the track elements can be assembled in a discontinuous fashion. However, the main drawback related to the discontinuous track comes from the edges of the magnet track, because the disturbance forces are higher than in the case where the mover is fully overlapping the magnet track. As a result, the edge-specific disturbance forces are increased cogging forces and electromechanical forces caused e.g. by circulating transient currents in the armature windings.

In this paper, the disturbance forces of a nonsalient permanent magnet linear motor with a discontinuous magnet track are studied. The motor under study, TB12S manufactured by Tecnotion, is modeled by a 2D FEM model, and the skewing of the magnets is taken into consideration by applying a slicing method. FEM calculations are used to analyze the cogging forces during edge crossing. An experimental test setup with two movers (linear motors) for the same permanent magnet track is used to demonstrate and validate the behavior of the cogging forces.

A. Discontinuous Magnet Track

At the edge of the track, the disturbance forces are higher than in a situation where the mover is totally overlapping the magnet track. The cogging forces occur as a function of position for a nonsaturated armature. This applies also at the edges of the magnet track. Naturally, the cogging forces are higher at the edges than in normal operation above the magnet plate, because some of the balancing forces are missing from that armature end, which is not anymore above the magnet plate. In Fig. 1, the principle of forces between the permanent magnets of the track and the iron teeth of the armature is illustrated. In Fig. 1 a), the interaction between the iron mover and the permanent magnets is depicted when the armature is fully above the magnet plate. The arrows in Fig. 1 represent the level and direction of the attraction force between the iron teeth and the magnets. It can be seen that there is a high attraction force to pull the iron mover towards the magnets, but there are also small horizontal forces. These horizontal forces cause the cogging forces. The force acting on each tooth is different in different mover positions, because the tooth position related to the positions of the magnet is different. However, in the case where the mover is fully overlapping the magnet track, some of the horizontal forces sum to zero, as shown in the green boxes in Fig. 1. The remaining horizontal force caused by the end teeth of the armature, and therefore, the design of the end teeth can be used for cogging force compensation.

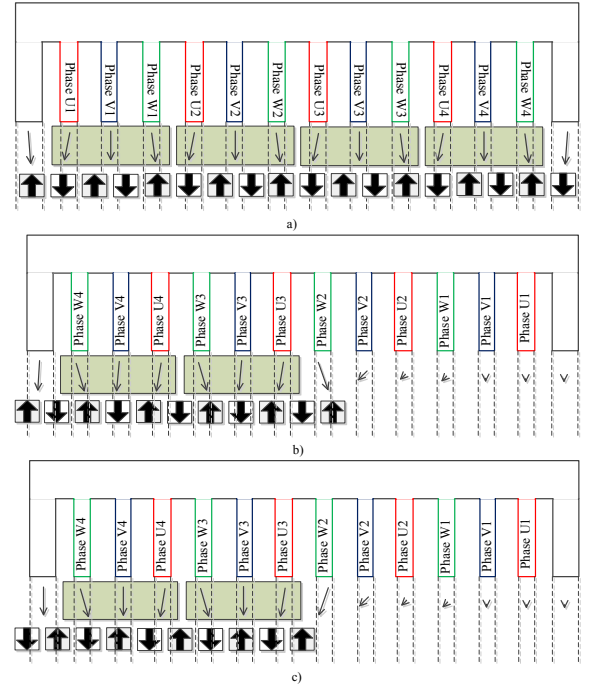


Fig. 1a) Principle of cogging forces when the armature is fully above the magnet plate, b) away from the magnet track, and c) towards the magnetic track.

At the edge of the magnet track the situation changes, because some of the symmetry is lost and the armature phase set at the edge does not have counteracting forces from different teeth of the phase set. Two different armature positions are shown in Fig. 1 b) and c). In Fig. 1 b), the sum of the horizontal forces is positive (away from the permanent magnet track), and correspondingly, in Fig. 1 c) the sum of the horizontal forces is negative (towards the permanent magnet track). It is pointed out that because there is some attraction between the last magnet of the magnet track and the teeth that have already exited from the magnet track, the average of the cogging force at the edge is not zero, because it has an offset in the negative direction.

B. FEM Modeling

Because of the ends of the mover, modeling of linear motors [12] differs from the modeling of rotating machines [13]. This means that boundary conditions satisfying adequately the flux behavior at the edges are needed, and the usage of symmetry-based calculation is more limited than with rotating machines. In this paper, Flux by Cedrat is used for FEM modeling, and the boundary condition issue is solved by providing a large air region with an infinite area around it as shown in Fig. 2 a). To reduce the calculation time in transient simulations, compressible areas at the ends of the track are used to make the movement of the motor possible. In these areas, the meshing is updated at each time step to match the current position while the rest of the mesh remains constant. In Fig. 2 b), a schematic of the linear motor FEM model is presented. As it can be seen, the linear motor is modeled as a 2D construction at the cross-

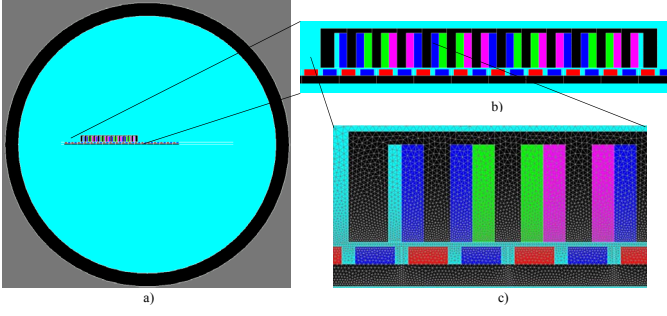


Fig. 2a) FEM model for the linear PM motor inside the air region with an infinite area; b) with the PM track and the end teeth, and c) example of the meshing of the linear PM motor in the FEM model.

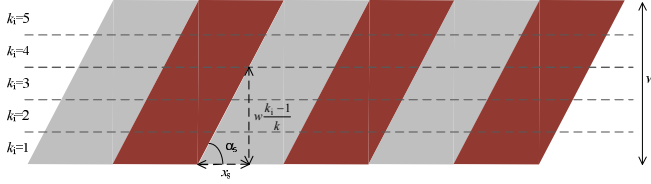


Fig. 3. Principle of the sliced PM track with skewed magnets.

section of the linear motor. The same winding construction is repeated four times in the armature and the four winding groups are connected in parallel. This winding arrangement allows circulating compensating currents run in the winding during overlapping the PM edge. In Fig. 2 c), the meshing of the FEM linear motor model is illustrated.

C. Skewing by the slicing method

In this paper, the whole motor with a discontinuous magnet track is modeled by a 2D FEM model, whereas the actual motor also has a third dimension. This means that in the 2D model only a cross-section of the motor is considered, and the motor structure is assumed constant towards the third dimension. However, the skewing of the magnets has a direct impact on the forces in the motor and cannot thus be neglected. Hence, the effects of skewing are considered in the modeling and estimated by the slicing method. The skewing angle of the magnets is known from the structure of the motor. The cross-section results achieved with the 2D FEM model can be used to estimate the effects of skewing. It can be assumed that the motor is divided into k parts in the direction of the width of the motor as shown in Fig. 3. As can be seen in Fig. 3, each part is position shifted. The position shift x_i of the k_i th part of the motor with a width of w is defined from the skewing angle α_s as

$$x_s(k_i) = \frac{w \frac{k_i - 1}{k}}{\tan(\alpha_s)} \quad (1)$$

In practice, the 2D model results for force could be calculated for all parts. However, the cross-section for all k parts is the same, except for the mover position shift, which is defined by Equation 1. Thus, the produced force can be estimated as a function of position x as

$$\hat{F}(x) = \frac{\sum_{k_i=1}^k F_{\text{shift}}}{k} \quad (2)$$

where F_{shift} equals the force of the 2D model in the position $x + x_s(k_i)$. In this paper, the motor skewing is taken into account by using five displaced motor segments.

III. ESTIMATION OF DISTURBANCE FORCES

An experimental test setup depicted in Fig. 4 with two movers (linear motors) for the same permanent magnet track is considered in order to analyze the behavior of the cogging and electromechanical forces during the edge crossings. The magnet track is modular, which allows free assembly of the magnet plates. Moreover, the active mover is connected electrically to the frequency inverter, and its current can be controlled while the passive mover electrical connection is left open.

The main idea of using the controlled active motor is to produce a constant velocity for the passive mover during the edge crossings. For this reason, the motion control topology of the active mover is a position cascade that consists of a P-type position controller $C_P(z)$ cascaded with a PI-type velocity controller $C_{PI}(z)$ combined with a velocity feedforward term K_{ffv} . The measured signals from the active mover are used to estimate the cogging forces when the passive mover is crossing the magnet track edge. During the crossing, also the currents of the armature windings are measured from the passive mover (see Fig. 4 and Fig. 5) in order to estimate the forces caused by the circulating winding currents. It is pointed out that because the distance between the active and passive movers is such that they are both magnetically in the same position, the position measurement of the active mover can be used for the d- and q- axis conversion. This provides an opportunity to use the q-axis current to estimate the force component.

A. Indirect estimation of cogging forces

The magnetic field power is a function of the permanent magnet flux linkage and the winding current and can be presented for one phase

$$P_{\text{field}} = \frac{d\Psi_{\text{magnet}}}{dt} i_{\text{ph}} = \frac{d\Psi_{\text{magnet}}}{dx} \frac{dx}{dt} i_{\text{ph}}, \quad (3)$$

where Ψ_{magnet} is the flux linkage of the permanent magnet and i_{ph} is the current of one phase. If the losses of the conversion have been neglected the mechanical output power has to be equal with the power in the magnetic field,

$$F \frac{dx}{dt} = \frac{d\Psi_{\text{magnet}}}{dx} \frac{dx}{dt} i_{\text{ph}}, \quad (4)$$

This form gives the possibility to express the force F as

$$F = \frac{d\Psi_{\text{magnet}}}{dx} i_{\text{ph}}, \quad (5)$$

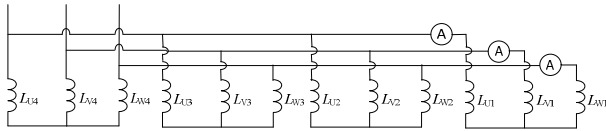


Fig. 5. Current measurement arrangement of the passive mover. Internal currents of all three phases from the first of the four parallel phase sets can be measured.

B. Electrical circuit

In the FEM model, an ideal current supply with a zero current reference for all coils is used. However, in the experimental verification, an actual motor is used that contains an electric circuit with four parallel three-phase winding sets as shown in Fig. 5. The back emf of each of these sets is, in turn, reduced in the time period when the winding is moving away from the magnet track area. As a result, balancing currents are running in the parallel branches of the winding for a moment, and, in addition to the cogging force, there is an additional electromechanical force, although no voltages or currents are supplied to the motor terminals of the passive motor. In Section IV, the forces caused by the circulating winding currents during the edge crossing are analyzed.

IV. RESULTS

The test setup comprises two Tecnotion TB12S linear motors connected together with a stiff connection as shown in Fig. 6. For the active mover, an industrial frequency converter acting as a PWM voltage source is controlled by a dSPACE DS1104 controller board. The magnetic pole pitch of the motor is 12 mm, and the tooth pitch between the teeth of two successive phases is 16 mm. The magnet track consists of several magnet plates with a length of 0.288 m.

Experimental tests are performed to validate the FEM model results. However, in dynamic operation, the direct measurement of the cogging force is difficult, because in practice, it would require either the measurement of the flux distribution from the short air gap of the mover during motion, or all other disturbance forces affecting the mover should be known so that the cogging forces could be estimated from the motion of the mover. With a rotating motor it usually suffices to measure the behavior of the flux over one pole pair with a fixed measurement connection, because the rest of the motor repeats that behavior [11]. However, it is pointed out that in

the case of a linear motor, the flux measurement area has to be either equal to the length of the moving distance, or the measurement sensors have to move along the mover because the linear motor does not return mechanically to the same position during motion as the rotating motor does after one full turn. Thus, the distance between the movers is essential. By adjusting the distance between the movers with an even multiple of pole pitch, the movers are magnetically in the same position at the magnet track. With such a configuration, the same cogging forces are affecting both of the movers, and the total cogging force is double when compared with the cogging forces of a single mover. By adjusting the distance between the movers differently, the total cogging forces can be decreased. If the distance between the movers equals the multiple of pole pitch, the movers are magnetically in opposite positions with respect to the cogging forces.

The active mover overlaps the magnet track all the time when the passive mover crosses the edge of the magnet track. Thus, the force constant of the active mover remains constant during crossing of the magnet track edge of the passive mover. Hence, the cogging force can be estimated by the method described above also at the edge of the magnet track by recording the current measured from the active mover.

A. Estimated Cogging Forces

The cogging forces have been determined during edge crossing with the constant velocity-based estimation routine described in Section III and compared with the FEM-modeled ones. In the experimental tests, a low constant velocity 0.15 m/s is considered and carried out so that one of the magnet plates has been removed from the track. To further validate the behavior of the cogging forces, in the second test, the track has been modified by replacing the removed magnet plate with an iron plate in order to change the air gap dynamics. In Fig. 7 the estimated cogging forces for both cases are compared with the FEM-modeled one (without magnet plate). Evidently, the estimated cogging forces for both assemblies are in a good agreement with the FEM-modeled one indicating that the estimation routine is valid. It can be seen that the shape of the force as a function of position has similarities, even though the level of the force has some differences.

The most noticeable difference is that the FEM model results seem to give higher cogging forces in the positive direction than the estimation by measurements. In the case of a positive cogging force it is accelerating the motor and the position controller has to brake the motor to correct the position. Friction force is always a braking force, and thus, in the case of a positive cogging force, the controller does not have to produce as much force as the cogging force actually is, and for this reason, the estimate remains too low. Nevertheless, the estimation yields a good approximation of the cogging force behavior at the edge of the PM track. It is important to notice that manipulation of the air gap length, that is, inserting an iron plate into the magnet track, does not have a significant influence when compared with the other estimation result.

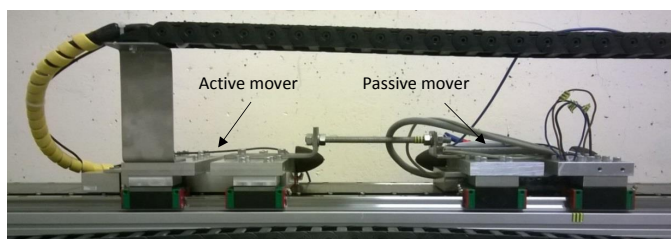


Fig. 6. Experimental test setup with two movers for estimating the cogging forces. The distance between the two movers is an even multiple of the pole pitch, and thus, the movers are magnetically in similar positions.

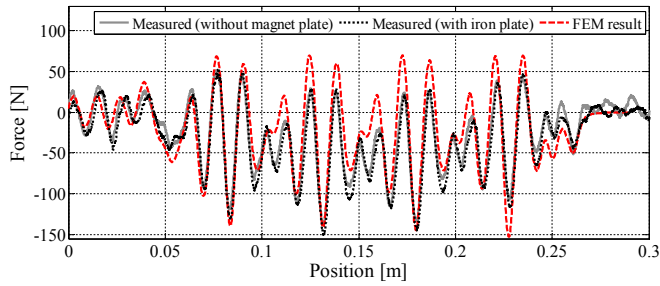


Fig. 7. Experimentally determined cogging force estimates when the magnet plate has been removed (gray line) and replaced by an iron plate (black dotted line) compared with the FEM-modeled one (red dashed line).

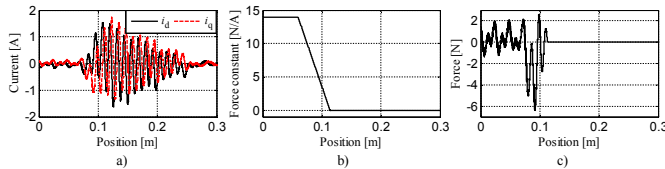


Fig. 8. a) Measured d- and q-axis currents from the passive mover when leaving the PM track, b) Motor force constant of the first phase set of the armature at the edge of the magnet track. After the position 0.108 m, the armature is not above the magnets and does not produce force. c) Force of the first phase set caused by internal currents at the magnet track edge.

B. Accuracy of the estimation routine

There are several factors that can cause differences between the measurement and FEM results, such as: disturbance forces of the other mover, friction as a braking force, controller bandwidth, electric circuit, and deviations from the constant velocity. Obviously, the bandwidth of the position controller limits the response to the disturbance forces, and thus, the estimated force does not fully match the actual disturbance force. More importantly, in the measurement, the active mover is used to produce the force needed for constant velocity, and the force estimate is based on the position controller output. Disturbance forces, such as friction and cogging forces, have a direct impact on the position error and thereby, on the controller output.

C. Electromechanical forces

When the passive motor crosses the edge, balancing currents are running in the parallel branches of the winding for a moment, and in addition to the cogging force, there is an additional electromechanical force, although no voltages or currents are supplied to the motor terminals. Fig. 8 a) depicts the q- and d-axis currents in the first phase set at the magnet track edge. All three phase currents are measured, and a dq transform is made. As it can be seen, there are high d-axis currents as the winding tries to compensate for the anomaly caused by the permanent magnet track edge. Again, Fig. 8 b) shows an approximation of the force constant of the first phase set. At the edge, the armature is exiting from the magnets, and the current in the armature winding does not produce a force any more. Because of that, the force constant is simply approximated to decrease linearly when the armature exits the

magnet track. The initial force constant is set to a value of one-fourth of the total force constant of the motor because the first phase set is producing one-fourth of the motor force with the motor with four parallel phase sets. The force produced by the winding currents at the magnet track edge is presented in Fig. 8 c). The result is achieved by multiplying the q-axis current of the first phase set in Fig. 5 with the force constant of the first phase set in Fig. 8 b). By comparing the results in Fig. 7 and Fig. 8 c), it can be seen that the forces caused by the circulating winding currents are small when compared with the cogging forces, and thus, it can be assumed that the accuracy of the estimation method is not affected by these forces too much.

D. Influence of friction

In this paper, it is assumed that the friction with a constant velocity is almost constant, and the resulting disturbance forces have been compensated for in the estimation by a feedforward controller. However, small variations in friction occur at the edge, and the friction force is proportional to the attraction force through a friction constant. These variations are caused by variations in the attraction force at the edge. The PM linear motor with a small air gap (1 mm in the test setup) has a high attraction force, but when the mover travels freely away from the magnet track, the attraction force is reduced from 3500 N to the value of the gravitational force. Consequently, also the friction force decreases as can be seen in Fig. 9, where the upper line with a constant value represents the average friction force with a constant velocity with two movers above the magnet track, and the lower line with a constant value indicates the average friction force with one mover above the magnet track. The difference is in the range of less than 10 N.

E. Constant velocity

In the FEM model calculation, a constant velocity of the armature is used, which is also assumed in the measurements. However, in the tests, the velocity is not totally constant because of the high cogging forces at the edge of the PM track. Obviously, deviations from the reference position and velocity are important in the sense that the cogging force estimation routine is based on the controller output, and without errors the force would be constant and thus, there would be no estimation results available. Naturally, some of the force is needed for accelerating and decelerating the armature. When the velocity errors remain small, also the required forces are small. By increasing the bandwidth of the position and velocity controllers, the velocity would remain closer to the velocity reference in the tests. However, there are certain practical limits for the controller bandwidth, such as delays in the measurements and control circuits and dead times needed for switching of the inverter semiconductors.

V. CONCLUSIONS

This paper studied a permanent magnet linear motor drive with a discontinuous magnet track and analyzed the cogging

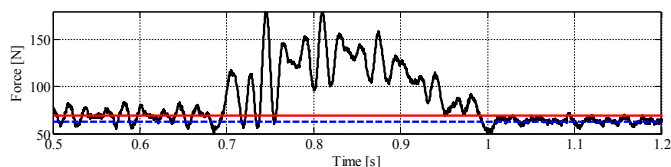


Fig. 9. Force estimation based on the position controller output. The friction force difference, between two movers (active and passive) on the magnetic track and one mover (active) on the track, can be seen from the average forces (upper and lower lines with a constant value). Mover velocity 0.15 m/s; in the interval 0.5–0.7 s the mover is on the magnet track; 0.7–1 s the mover is leaving the track, and 1–1.2 s is free motion.

forces at the edge of the magnet track. An estimation routine for the cogging forces at the edge of the magnet track was presented, which is based on an experimental test setup with two linear motor movers attached to each other. The estimated cogging forces were compared with the corresponding values obtained by a simplified FEM model, which was considered as a reference as it provides an accurate machine model but neglects the nonidealities of the experimental test setup. Evidently, the 3D FEM could provide more accurate results especially at the edge crossings which is a topic of future research. Also the analyzation by 3D FEM model could provides better indication of the influence of the skewing. However, the main objective of this paper was to study cogging forces during edge crossings and the results between the simulations and experimental results were in a good agreement.

Despite some differences between the results, such as in the vibration amplitudes and the frequency content, the values obtained by the FEM analysis show a good agreement with the measurements and validate the behavior of the cogging forces during the edge crossings. Causes for the differences in the measurements results were analyzed, and it was found that the main cause is nonlinear friction, which is neglected in the FEM model. Nevertheless, the experimentally determined cogging forces were in a satisfactory agreement with the FEM model at the edge crossings, which was also seen in the case of different track element assemblies; by removing a magnet plate from the track and by replacing the removed magnet plate with an iron plate.

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